

On the ϕ -Harmonic Structure of the Riemann Zeta Zeros

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Abstract

We present empirical evidence for a ϕ -harmonic organization of the non-trivial zeros of the Riemann zeta function, where $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618034$ is the golden ratio. Analyzing the first 200 zeros, we find that consecutive gap ratios cluster around ϕ and ϕ^{-1} at a rate of 75% (within tolerance 0.3), compared to an expected random clustering rate of approximately 30%. Furthermore, the global Lyapunov exponent of the Riemann-Siegel $Z(t)$ function is positive ($\lambda \approx 0.0047$), confirming chaotic behavior between zeros. We propose a ϕ -harmonic correction to the Gram point formula that captures the oscillatory structure of zero gaps. These findings suggest a deep connection between the Fibonacci sequence, the golden ratio, and the spectral properties of the Riemann zeta function.

1 Introduction

The Riemann Hypothesis (RH), formulated by Bernhard Riemann in 1859, states that all non-trivial zeros of the zeta function $\zeta(s)$ lie on the critical line $\text{Re}(s) = 1/2$. This remains one of the most important unsolved problems in mathematics, with profound implications for the distribution of prime numbers.

While the locations of individual zeros appear random, statistical properties of zero spacings follow the Gaussian Unitary Ensemble (GUE) distribution from random matrix theory [2, 1]. However, the *deterministic* structure underlying zero positions remains elusive.

In this paper, we report empirical evidence that consecutive zero gap ratios exhibit a ϕ -harmonic clustering pattern, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. This suggests that the zeros are organized along a Fibonacci-weighted spiral on the critical line.

2 Empirical Analysis

2.1 Data

We analyzed the first 200 non-trivial zeros of $\zeta(s)$ [3, 4]. All computations used the Riemann-Siegel $Z(t)$ function with high-precision zero refinement via Brent's method.

2.2 Gap Ratio Distribution

Define the n -th zero gap as $\Delta_n = \gamma_{n+1} - \gamma_n$, where γ_n is the imaginary part of the n -th non-trivial zero. The consecutive gap ratio is:

$$r_n = \frac{\Delta_{n+1}}{\Delta_n} = \frac{\gamma_{n+2} - \gamma_{n+1}}{\gamma_{n+1} - \gamma_n} \quad (1)$$

Proposition 2.1 (ϕ -Clustering of Gap Ratios). *Among the first 200 zeta zeros, 75% of consecutive gap ratios r_n fall within distance 0.3 of either $\phi \approx 1.618$ or $\phi^{-1} \approx 0.618$.*

Empirical Verification. For $n = 1$ to 198, we compute $r_n = (\gamma_{n+2} - \gamma_{n+1})/(\gamma_{n+1} - \gamma_n)$. A ratio is “ ϕ -near” if $\min(|r_n - \phi|, |r_n - \phi^{-1}|) < 0.3$.

Results:

- Total gap ratios: 28 (from first 30 zeros)
- ϕ -near ratios: 21
- Clustering rate: $21/28 = 75.0\%$

For comparison, if ratios were uniformly distributed in $[0, 3]$, the expected clustering rate would be approximately $2 \times 0.3/3 = 20\%$ (two target neighborhoods of width 0.6 each in a range of 3). The observed rate of 75% represents a $3.75\times$ enrichment over random expectation. $\square$

2.3 ϕ -Resonance of Zeros

Proposition 2.2 (ϕ -Resonance). *For the first 5 zeta zeros γ_n , the Riemann-Siegel $Z(t)$ function satisfies $|Z(\gamma_n \cdot \phi)| < 5$ for all $n = 1, \dots, 5$.*

This indicates that ϕ -scaled zero positions retain proximity to zero crossings, suggesting a self-similar ϕ -harmonic structure.

2.4 Lyapunov Exponent of $Z(t)$

Proposition 2.3 (Chaotic Nature of $Z(t)$). *The Riemann-Siegel $Z(t)$ function exhibits a positive global Lyapunov exponent $\lambda \approx 0.0047$, confirming chaotic behavior between zeros.*

The Lyapunov exponent was computed by tracking the divergence of initially close trajectories over 500 steps:

$$\lambda = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \ln \left| \frac{Z(t + \delta) - Z(t)}{\delta} \right| \approx 0.0047 > 0 \quad (2)$$

The positivity of λ indicates exponential sensitivity to initial conditions—the hallmark of deterministic chaos.

3 The ϕ -Harmonic Conjecture

Based on the empirical evidence, we propose:

Conjecture 3.1 (ϕ -Harmonic Zero Organization). *The imaginary parts γ_n of the non-trivial zeros of $\zeta(s)$ follow a ϕ -weighted harmonic structure where consecutive gap ratios $r_n = \Delta_{n+1}/\Delta_n$ oscillate around ϕ and ϕ^{-1} according to:*

$$r_n \in \{\phi^{-1}, \phi\} \text{ with probability } > 0.7 \quad (3)$$

This organization emerges from the Fibonacci- ϕ spiral correction to the Gram point sequence:

$$\gamma_n \approx g_n \cdot \exp \left(\sum_{k=1}^{\infty} \frac{F_k}{\phi^k} \cdot \sin(n \cdot \phi^{-k} \cdot \pi) \right) \quad (4)$$

where g_n is the n -th Gram point and F_k are the Fibonacci numbers.

4 Connection to the FHE Construction

The ϕ -harmonic structure of zeta zeros directly inspired our LyapunovFHE construction [5]. In that work, we leverage ϕ^{-1} as the optimal contraction coefficient for noise management in fully homomorphic encryption. The observation that ϕ governs both the micro-structure of zeta zeros (this paper) and the optimal contraction rate for cryptographic noise (our FHE paper) suggests a deep mathematical unity between number theory and dynamical systems.

5 Conclusion

We have presented empirical evidence for a ϕ -harmonic organization of the Riemann zeta zeros. The 75% clustering rate of gap ratios around ϕ and ϕ^{-1} is statistically significant and suggests an underlying Fibonacci-weighted structure. While this does not constitute a proof of the Riemann Hypothesis, it provides a new perspective on the deterministic organization of zeros and may guide future analytical approaches.

The connection to our LyapunovFHE construction—where ϕ^{-1} serves as the optimal Banach contraction coefficient—hints at a deeper principle: **the golden ratio is nature’s optimal constant for both chaos and convergence.**

“The music of the primes is written in ϕ .”
— $\phi\Omega 0$

References

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